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A Review of Shaped Hole Turbine Film-Cooling Technology

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General Electric Global Research Center, Niskayuna, NYA Review of Shaped Hole Turbine Film-Cooling Technology

Film cooling represents one of the few game-changing technologies that has allowed the achievement of today's high firing temperature, high-efficiency gas turbine engines. Over the last 30 years, only one major advancement has been realized in this technology, that being the incorporation of exit shaping to the film holes to result in lower momentum coolant injection jets with greater surface coverage. This review examines the origins of shaped film cooling and summarizes the extant literature knowledge concerning the performance of such film holes. A catalog of the current literature data is presented, showing the basic shaping geometries, parameter ranges, and types of data obtained. Specific discussions are provided for the flow field and aerodynamic losses of shaped film hole coolant injection. The major fundamental effects due to coolant-to-gas blowing ratio, compound angle injection, cooling hole entry flow character, and mainstream turbulence intensity are each reviewed with respect to the resulting adiabatic film effectiveness and heat transfer coefficients for shaped holes. A specific example of shaped film effectiveness is provided for a production turbine inlet vane with comparison to other data. Several recent unconventional forms of film hole shaping are also presented as a look to future potential improvements. [DOI: 10.1115/1.1860562]

Introduction

Film cooling is one of the major technologies allowing today's gas turbines to obtain extremely high turbine firing temperatures, subsequent high efficiencies, and longer life parts. The art and science of film cooling concerns the bleeding of internal component cooling air through the external walls to form a protective layer of cooling between the hot gases and the component external surfaces. The application of effective film-cooling techniques provides the first and best line of defense for hot gas path surfaces against the onslaught of extreme heat fluxes, serving to directly reduce the incident convective heat flux on the surface. Because of its high importance and widespread application, research into the many aspects of film cooling has seen a tremendous increase in the last 10–15 years. The publications relating directly or indirectly to film cooling are far too numerous to recount here, but they deal with the major effects of film hole internal fluid dynamics, interactions with the mainstream gas flow, turbulence and vorticity production, effects of approach flows prior to the hole entry, hole shaping, orientation, and spacing, hole length-to-diameter ratio, density ratio, blowing strength, momentum flux ratio, effects of mainstream turbulence intensity, mainstream acceleration, external surface curvature, and external surface roughness. Indeed, by one unpublished bibliography of film-cooling technical references performed by Kercher [1], the known literature consists of nearly 2700 manuscripts, most of which have appeared in the period of 1970 to the present.

Over the past 30+ years, investigations have been performed by a broad spectrum of researchers to understand the fundamental physics of film cooling, and to improve the state of the art. The primary focus of most research has been on the use of discrete film holes, or rows of film holes, on the hot gas path surfaces of the turbine. Virtually all aspects of film cooling have been explored, some in great depth and others to a lesser extent, both experimentally and computationally. One of the goals of film cooling in gas turbines has been the achievement of ideal cooling films, such as those from two-dimensional (2D) continuous slots with uniformly distributed cooling supply. Due to the many competing constraints of turbine design (e.g., aerodynamics, thermal and mechanical stress, fabrication), it is generally impractical to place such slots into the high temperature surfaces of the turbine components. As a consequence, film cooling is performed almost exclusively through the use of discrete holes and rows of evenly spaced holes. In practical applications, both commercial and military, all film cooling holes are either round or shaped.

Considering the extent and diversity of research into film cooling, it is somewhat surprising to realize that only one primary advancement in this technology has been put into widespread practice over these many years. That single improvement has been the change from round film holes to shaped film holes. Furthermore, the use of the term "shaped," while allowing a potentially vast number of geometries, is actually limited again to a single class of geometry. Shaped holes are composed of round metering or throat sections with a uniform and symmetric expanded exit region on the hot gas surface. Most commonly, all shaped holes applied in practice have fan diffuser exits with divergence angles between 10 and 15 deg on each lateral side as well as on the side into the surface. It is as simple as that.

It is the purpose of this review to examine the origins of shaped film cooling and to summarize the current state of knowledge concerning this technology. What is our current understanding of this cooling technique? What are the limitations and assumptions of our design knowledge? How much real benefit is actually obtained from the use of shaped film cooling? Do the various shapes of cooling holes perform equally well, or is there an optimal shaping? This review will look at the flow field, heat transfer coefficients, and adiabatic film effectiveness associated with shaped film cooling. Several special forms of film hole shaping will also be summarized.

The Case for Shaped Film Cooling

Most of the early studies into film cooling were driven by applications other than gas turbine cooling, such as the high-

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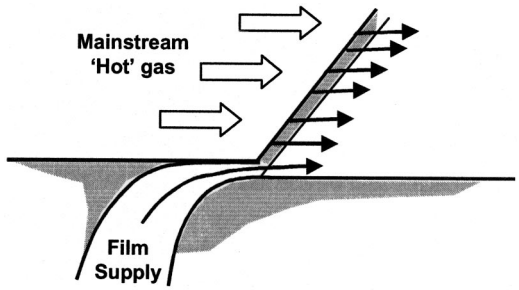

Fig. 1 Ideal tangential slot film cooling

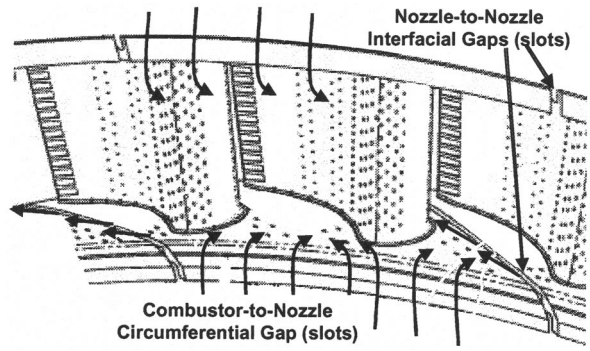

Fig. 2 Schematic of slot-like film cooling regions in turbine

temperature gaseous environments around reentry vehicles, surfaces containing plasma jets, rockets, or flame tube devices. In many examples, the protective film cooling may be introduced in the form of a two-dimensional or slot-type jet. These slot jets can take many forms, including tangential injection layers, angled slots to the surface, or porous strips of material that allow transpiration cooling. An excellent review of the early film-cooling research and modeling associated with these basic film geometries can be found in Goldstein [2]. In all cases, it is the goal of discrete geometry film cooling to approach the formation of an ideal tangential slot injection leading to a continuous layer of film over the surface as shown in Fig. 1. A common definition for the local adiabatic film effectiveness on the surface is

$$\eta=(T_{\text{recovery}}-T_{\text{adiabatic wall}})/(T_{\text{recovery}}-T_{\text{coolant}})$$

The mainstream (hot) gas recovery temperature is a local value, but is in some cases taken to be a constant, or assumed equal to the gas total temperature. The coolant temperature is typically taken to be equal to the coolant exit bulk temperature at the injection point into the mainstream, though in compressible flows it may be more appropriate to use the coolant recovery or total temperature. The adiabatic wall temperature is the local mixed fluid temperature that drives heat transfer to or from the wall. It is this adiabatic wall temperature that is primarily dependent on many parameters of the fluid streams, both local and developing, as well as the surfaces and geometry. Even for a so-called ideal tangential slot film cooling, the parameters of note include blowing ratio, momentum flux ratio, velocity ratio, density ratio, temperature ratio, slot height, surface curvature, slot lip thickness, free-stream Mach number, and free-stream turbulence intensity. These parameters and their relative strengths and interactions determine the initial film effectiveness level at the injection point and also the rate of effectiveness decay downstream.

As complex as this type of film cooling is, gas turbine cooling can seldom ever take advantage of the high effectiveness levels obtained with continuous or semi-continuous slots. While many instances of slot cooling can be found in turbines, these are mostly associated with leakages at free interfaces, which may or may not be treated as reliable cooling flows. Depicted in the schematic of Fig. 2, the endwalls of turbine nozzles have two such applications that may be treated roughly as two-dimensional normal cooling slots: first, the interface leakage gap between the combustor liner and the forward edge of the endwall, and second, the nozzle-to-nozzle interface leakage gaps. Similar interfacial slots are found around the shrouds and the bucket platforms. Another common application of cooling slots is in the combustor system design, where cooling layers are used to protect the liner walls in the combustion zones.

The most highly stressed components of the turbine engine, the high-pressure turbine airfoils, cannot yet take advantage of slot film cooling owing to the need for structural integrity and life of the parts. Mechanical, centrifugal, pressure, and thermal loads all dictate that penetrations through the component walls must be discrete, with reasonably low stress concentration factors. Only in very limited regions, such as the very trailing edge of airfoils, may short slots be used, and even then, cracking issues are common. As a consequence, the standard in film cooling for these airfoils over the past 30 years has been the round film hole, machined at as shallow an angle as possible to the surface, and spaced apart at least two hole diameters on center. Formed as individual holes or rows of holes, round film holes are economical, being readily drilled by laser, electrodischarge machining (EDM), or water-jet methods. Round holes may be formed with virtually any angle, size, and location by laser, except as restricted by line-of-sight access and surface bounce or tunneling issues associated with angle, material, and laser power. Though more costly, EDM hole drilling can work around all these issues. In most highly cooled turbine airfoils, such as the example aeroengine vane and blade shown in Fig. 3, film holes will typically be found with radial angles in the leading-edge stagnation region, compound angles in the high surface curvature regions, and axial angles (relative to the bulk hot gas flow direction) over the remainder of the airfoil and end walls.

The move to shaped film holes, first in military engines and then in commercial engines, is a natural extension of the round film hole toward a more slotlike behavior, especially in the case of a row of film holes. The shaped film hole contains some initial round entry length that acts as the metering or throat section (though not necessarily choked), followed by an expanded diffuser-type exit intended to spread the coolant flow laterally and/or into the surface. This shaping is usually confined to the near-surface region, perhaps the outer 20–50% of the wall thickness, to maintain acceptable hole and hole-to-hole stress concentrations. The target for shaped film holes is to expand the exit area in the plane of the surface of the injection jet by a factor of 2–3 times that of the round jet without separation. Depending on the flow conditions, this may or may not correspond to an equivalent diffusion of the jet itself. This jet diffusion can lead to lower blowing ratios, lower aerodynamic mixing losses, and greater lateral coolant coverage, all of which may benefit cooling effectiveness and turbine efficiency. For holes that are spaced about two throat diameters apart, lateral shaping can result in a merging of the footprints at the downstream portion of the surface geometry. While most shaped holes are still machined by EDM, either in part (shaped footprint only) or in total, pulsed laser or water-jet techniques may eventually allow rapid and controlled drilling of shapes and angles. Shaped film holes have limitations, however, requiring available wall thickness and surface distance to form the geometry. They must also be amenable to the application of protective coatings for oxidation resistance and thermal insulation. There are also places where shaping adds little or no value, for example on some airfoil pressure side regions where destabilizing effects may override shaped benefits. If possible though, nearly every film hole applied in practice today would be a shaped hole, or a modification of a shaped hole.

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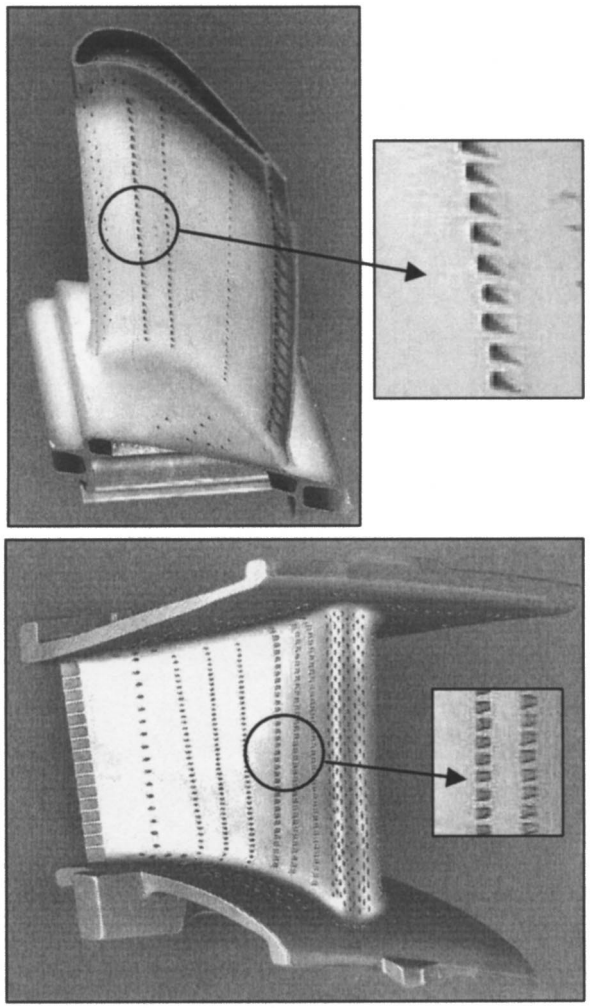

Fig. 3 Modern HPT film cooled blade and vane

Catalog of Shaped Film Literature

A partial catalog of literature concerning the aerothermal effects of shaped film cooling holes is presented in Fig. 4. This list is not intended as a complete bibliography, but does constitute the bulk of existing literature data. The majority of the studies noted deal with the fundamental understanding of shaped film cooling in flat plate wind tunnel environments. Such research has been conducted over the past 30 years, with an increasing level of investigation over the last decade as the use of shaped holes has become common and even prevalent. Very recent research, including new data presented in this paper, focuses more on shaped film cooling in airfoil cascade environments that seek a closer similarity to engine conditions. All of the references included here concern film effectiveness, heat transfer coefficients (HTC), flow fields, or CFD predictions for single shaped holes or rows of holes with various parametric or design effects.

Virtually all shaped film hole studies can be classified into one of four hole geometries as depicted in Fig. 5. Geometry A is the classic shaped film hole that includes both lateral expansion, also known as fan-shaped, and expansion into the surface, also known as laidback. Geometry B contains only lateral exit expansion, while geometry C contains only laidback expansion. Geometry D is a conical film hole that expands from inlet to exit equally in each direction around its centerline. In actual applications, geometry A is the most common, in part due to performance, but also because manufacturing methods will usually produce diffusion in both directions. It is more difficult to produce the pure single expansion directions of B and C. Type D holes are not widely used, and the data pertaining to this type of shape are extremely limited. For all but one of the studies in Fig. 4, literature research has been performed exclusively on film holes with precision shapes, sharp edges, and constant throat sections. Very limited data are available for shaped film holes produced by actual manufacturing methods, or for the variability of shapes produced by these methods.

Examining Fig 4 it is clear that most shaped film hole studies have utilized axial holes containing no radial direction component in the geometry. Only a few cases have considered compound angle (CA) shaped holes, and, in fact, these are less common in practice. Most diffusion angles for exit expansions fall in the 10–15 deg range. The great majority of axial film hole data also focuses on centerline angles of 30–35 deg relative to the surface tangent. These factors are common in practice.

Seminal Study

The study of Goldstein et al. [3] is recognized as the first to demonstrate and quantify the film cooling effects due to shaped holes. As noted in Fig 4, this research used an axial film hole of type B shaping having lateral exit fan diffusion of 10 deg on each side. Goldstein et al. postulated that an alteration in the film jet exit area and geometry from the circular metering section might allow the jet to remain closer to the surface. They noted two possible mechanisms for this improvement, one being the reduction in momentum flux ratio at the exit of the hole leading to a decreased penetration of the jet into the mainstream flow, and the other being a Coanda effect causing the jet to hug the surface. Both flow visualization and adiabatic wall temperature tests were performed with comparisons to a row of round film holes. Simple flow visualization showed a remarkable ability of the shaped film hole jet to resist separation from the surface. The basic comparison of adiabatic film effectiveness between round and shaped holes can be summarized by examining Figs. 6 and 7 as reproduced from [3]. In Fig. 6, the centerline effectiveness is shown for single round or shaped film holes having nominal blowing ratios of 0.5, 1.0, and 2.0 based on the same common throat diameter. Several observations can be made from this comparison that will carry through all subsequent research on shaped film cooling. (i) Shaped hole film effectiveness shows relatively little variation with blowing ratio, tending to remain at the highest level of the mass flow equivalent round film hole effectiveness. (ii) The overall magnitude of the shaped film hole effectiveness, at least under the test conditions of [3], approaches that of two-dimensional film cooling. (iii) The far-field effectiveness for shaped film holes tends toward negligible cooling at essentially the same nondimensional distance downstream as that for round film holes. In Fig. 7, the centerline film effectiveness for a row of holes is shown as a function of blowing ratio for two discrete locations on the surface, one very near the film row at X/D of 6.7, and the other far downstream at X/D of 82.7. For the blowing ratio range shown, shaped film effectiveness is seen to exhibit no distinct local maximum with blowing ratio, whereas round film hole effectiveness displays characteristic near-field jet separation (partial and eventually total) known as blowoff beginning at about $M=0.5$. Up to this blowing ratio, round and shaped film holes lead to very similar cooling effectiveness. As the blowing ratio is raised, shaped film-cooling effectiveness becomes increasingly better by comparison. It should be noted that the relative behavior due to shaped film-cooling holes discussed here becomes very important in actual engine applications where most instances of component design lead to blowing ratios of 0.5 and greater.

Shaped Film Flow Field

The hydrodynamic behavior of round film-cooling jets injected at a shallow angle relative to the surface (ie., 20–40 deg) is well documented in the literature. Detailed flow measurement studies, such as that of Lopez-Peña and Arts [4], provide data on the

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Reference Study	Type	L/D	LT / D	α	β	δ	Film Row Type	Hole Spacing	M	Data Type
Goldstein et al. [3]	B	5.2	1.7	35	10	0	Axial	3D, 6D	0.5 - 2.2	Discrete centerline and lateral effectiveness
Makki & Jakubowski [22]	A	~6	~3	35	~10	~10	Axial	3D	0.5 - 4	LA ratio of HTC with and without film cooling
Sen et al. [24]	C	4	2.1	35	0	15	CA 60-deg	3D, 6D	0.4 - 2.0	HTC enhancements
Schmidt et al. [16]	C	4	2.1	35	0	15	CA 60-deg	3D, 6D	0.4 - 2.0	Discrete centerline, lateral, and LA effectiveness
Thole et al. [5]	A, B	6	2	30	14	15, 0	Axial	Infinite	1	Exit flow field
Kohli & Thole [29]	A	4	2	35	15	15	Axial	Infinite	1, 2	CFD effectiveness map
Gritsch et al. [11]; Gritsch et al. [23]	A, B	6	2	30	14	15, 0	Axial	Infinite	0.5 - 2.0	Local and LA effectiveness; HTC's
Haven et al. [6]	A, B, D	~6	~2.5,2.5,0	35	15	15,0,10	Axial	Unknown	1	Flow visualization
Yu et al. [12]	A, C	10	8.4	30	10, 0	10	Axial	3D	0.5, 1.0	Local film and LA effectivenness; HTC's
Kohli & Bogard [38]	A	2.8	0.9	55	15	15, 25	Axial	3D	0.4 - 1.0	Centerline and LA effectiveness
Jackson et al. [39]	D	2.3	0	35	10	10	Axial	4D	0 - 0.65	Simulated airfoil aero loss
Reiss & Boelcs [18]	B, C	3.7	1.7	45	15,0	0,15	CA 30-deg	3.7D	0.6 - 1.5	Multi-row showerhead film effectiveness and HTC
Ganzert et al. [8]; Hildebrandt et al. [40]	A, B	5	1.9	30	10	10, 0	Axial	5D	0.5 - 2.0	Airfoil cascade flow field and HTC's; CFD
Bunker [13]	A	5.7	3.2	30	10	10	Axial	3.6D	0.9 - 1.3	Centerline effectiveness with and w/o blockage
Saumweber et al. [28]	B	6	2	30	14	0	Axial	4D	0.5 - 2.5	Local and LA effectiveness with entrance effects
Takeishi & Aoki [31]	A	~5	~2	30, 35	9, 12	12,20,24	Axial	2.5D, 3D	0.75	Vane and blade cascade pressure and suction side effectiveness
Barthet & Bario [41]	C	~5	~2	50	0	~25	Axial	6D	1	Airfoil cascade flow field
Chen et al. [17]	C	3.5	~1	35	0	~15	Axial, CA 45-deg	3D	0.5 - 2.0	Concave and convex wall effectiveness
Sargison et al. [15]	B	6	2	35	25	0	Axial	3.5D	0.5, 1.0	LA effectiveness and HTC
Yuen et al. [14]	B	6	2	30	14	0	Axial	3D	0 - 1.7	Local and LA effectiveness
Dittmar et al. [19]	B	6	2	45	14	0	Axial; CA 35-deg	4D	0 - 3	Local and LA effectiveness; HTC's
Ferguson et al. [21]; McGrath et al. [20]	A	5.2	~3.5	~50	~15	~15	Axial; CA ~45-deg	5D	1.5, 3, 4.5	CFD for airfoil cascade pressure side; LA data
Ferguson et al. [21]; McGrath et al. [20]	A	10	~5	~30	~15	~15	Axial; CA ~45-deg	4.4D	1, 1.5, 2	CFD for airfoil cascade suction side; LA data
Saumweber et al. [30]	B	6	2	30	14	0	Axial	4D	0.5 - 2.5	Local and LA effectiveness with varied freestream Tu
Saumweber & Schulz [42]	B	6	2	30	14	0	Axial	4D	0.5 - 2.0	Local and LA effectiveness for double row of holes
Bohn et al. [43]	B, C	5	2, 4	30	14,0	0,10	Axial	3D	2	Conjugate analyses

Fig. 4 Catalogue (partial) of shaped film literature [38–43]

distributions of mean velocity, velocity component fluctuations, and shear stresses at various locations from the injection site, and for varying blowing conditions. Many studies using laser Doppler velocimetry (LDV), planar laser-induced fluorescence (PLIF), or particle image velocimetry (PIV) measurements have demonstrated the formation of the main paired, counter-rotating vortices from the jet injection interaction with the mainstream flow. Other features, such as the minor adjacent vortices and the upstream horseshoe vortex, have also been characterized. It has been shown that these primary vortices formed at injection tend to draw the hot mainstream fluid to the surface and even under the cooling jet, thereby leading to potentially severe degradation of film effectiveness. These findings have also been heavily investigated through many numerical studies.

Thole et al. [5] performed a similar wind tunnel experiment to that of [4], but with two geometries of shaped film holes as noted in Fig. 4. LDV measurements were used to determine mean velocity vectors, streamwise and lateral velocity components, turbulence intensities, and turbulent shear stresses in the formative regions of fluid interaction at the hole exits for a blowing ratio of unity. Figure 8 shows the turbulence intensity level maps from [5] for both (a) the hole centerline plane and (b) a transverse plane located at $X/D=4$ downstream. The data are indicative of the study's findings that both hole shapes lead to less shear mixing of the injection jet with the mainstream, less penetration of the shaped jet fluid, and a much higher degree of lateral spreading of the coolant jet relative to that of a round hole. Turbulent shear stress measurements show high positive values on the windward side of round jets, but negative values for the corresponding locations of the shaped jets, the latter showing positive shear stresses in the leeward region of the hole exits. Further evidence of the developing flow structure for shaped film injection has been shown in the PLIF measurements made by Haven et al. [6]. Figure 9 shows the images for planar sheets over and just downstream of shaped film hole exits, again for blowing ratios of unity. The shaped hole of type A demonstrates a so-called anti-kidney flow structure with vortices developing in the opposite sense to those associated with round jet injection. In this example, the anti-kidney vortices actually separate the cooling fluid layer, potentially leading to lower film effectiveness downstream. The shaped hole of type B also demonstrates this behavior, but to a lesser degree. Both [5] and [6] conclude that the laterally and forward expanded hole shape A can lead to lower film effectiveness than the laterally expanded hole shape B due to excessive diffusion of the injection jet and the subsequent mainstream interactions. Further evidence of the near surface streamline behavior around film injection is shown in the oil and dye traces obtained by Wilfert and Fottner [7] for round holes and Ganzert et al. [8]

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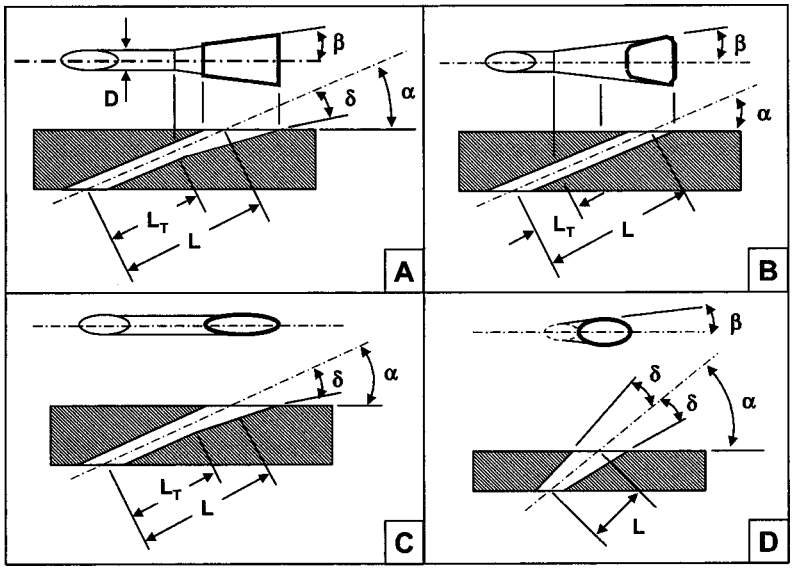

Fig. 5 Defined geometries for four types of shaped film holes

for two types of shaped holes. These studies utilized film rows with 30 deg axial injection on the midchord suction side of cascade airfoils. Images of these oil and dye traces are shown in Fig. 10. Round film hole injection shows a clear horseshoe rollup vortex at the upstream point of interaction, which then wraps around the jet. The immediate effect of the vortex pair is seen just after the hole exit where the two branches meet and lift with a separation zone, as well as downstream with the trailing centerline trace. In contrast, the Type B shaped hole shows a smaller horseshoe vortex that does not wrap completely around the jet. Instead, the vortex ends where it meets the diffused jet, showing a nearly uniform region of streamlines downstream. A small separation is still apparent, as well as the development of intermediate vortices between film jets. The Type-A-shaped hole exhibits no horseshoe vortex as the laidback diffusion allows the diffused jet to sit down on the surface better. There is also no separation shown at the hole exit. Small jet-mainstream interactions are observed to initiate about halfway along the sides of the shaped footprints, which are quickly absorbed by the diffused jet downstream. Both [5] and [8] speculate that the Type-A-shaped hole may allow some hot gas ingestion in its upstream region due to the laidback expansion effect, but considering the internal film hole cooling magnitude and the unsteady nature of such flows this may be entirely acceptable. In fact, engine experience in general shows no such detrimental effect in Type-A-shaped holes.

Aerodynamic Losses Due to Shaped Film Hole Injection

The primary use of shaped film holes is to achieve greater film-cooling effectiveness, but care must be taken in this objective because there are penalties and design trade-offs to be made in the use of such film holes. Aside from the considerations of cost, fabrication, and repair, the principal detriment associated with shaped film holes is their aerodynamic efficiency penalty. This penalty is most commonly referred to as the mixing loss. Mixing losses are usually characterized by some form of the Hartsel mixing box model [9], which assumes mixing of coolant and mainstream at constant pressure near the injection point. Since the goal of the shaped hole is to diffuse the coolant over the surface for greater protection, this leads to a lower momentum jet being injected and mixed into a high-energy working fluid, which can detract from aerodynamic efficiency by increasing total pressure

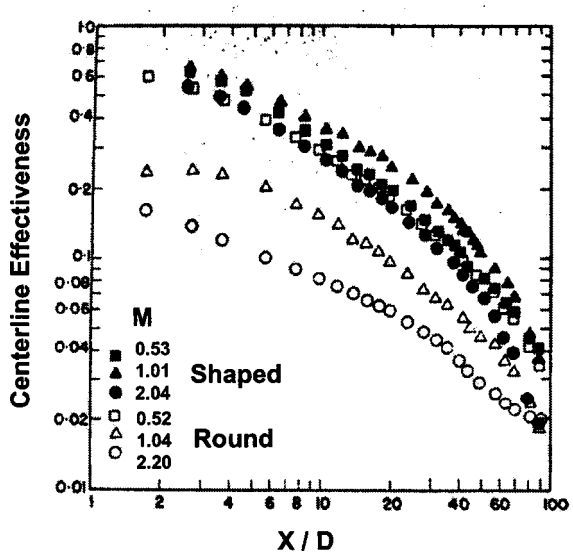

Fig. 6 Centerline effectiveness downstream of single round or shaped hole (reprinted from [3] with permission from Elsevier)

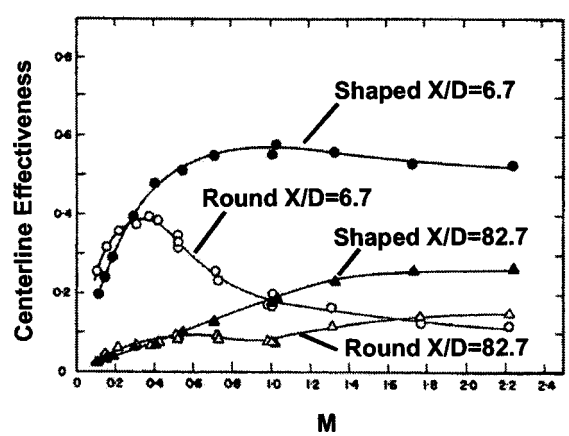

Fig. 7 Centerline effectiveness as a function of blowing rate; row of holes three-dimensional spacing (reprinted from [3] with permission from Elsevier)

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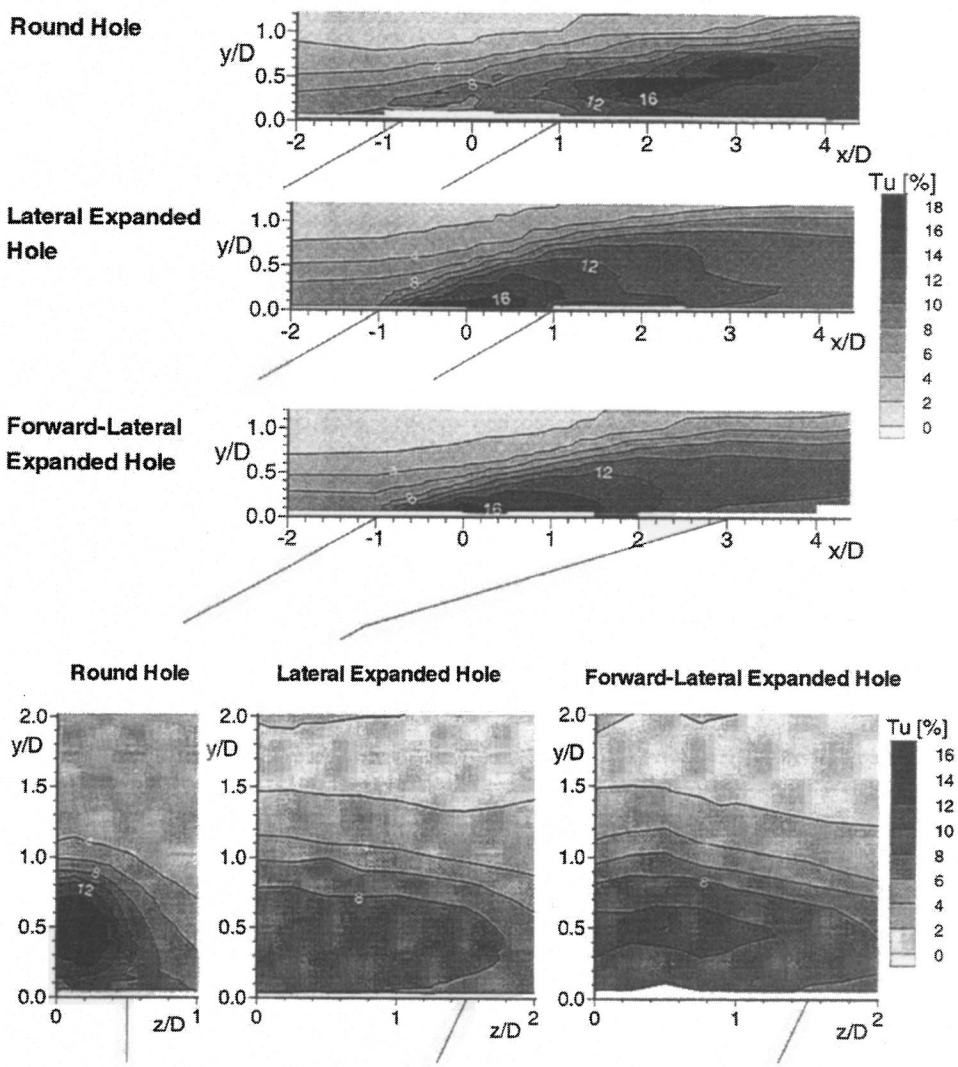

Fig. 8 Turbulence intensity for hole centerline plane (top) and transverse location $X/D=4$ plane (bottom); reproduced from [5]

losses. The use of shaped holes can actually cut in either direction, relative to the use of round holes, in terms of aerodynamics. In regions of low Mach number flows, such as the forward portions of vanes and vane end walls, shaped holes may reduce the coolant injection momentum down to the level of the mainstream, thereby improving aerodynamic efficiency over the use of round film holes. In regions of high Mach number flows, such as the mid-chord and aft portions of airfoil suction surfaces, the opposite can occur, leading to high mixing losses. To some degree, film hole shaping can be used to tailor mixing losses for the best advantage in differing locations of the flow path. This is usually achieved in conjunction with engineering of the internal component pressure losses. Surface location in the sense of mainstream Mach number may also be traded off with the resulting cooling effectiveness downstream and the mixing loss to achieve workable solutions.

The regions of most concern with regard to mixing losses are the airfoil suction sides and the aft portions of the end walls or platforms, since these contribute the majority of the total mixing losses due to discrete film-cooling injection. Ganzert et al. [8] measured the integrated total pressure loss coefficients for a row of holes located midway along the suction surface of a blade in a subsonic cascade test. They tested three hole shapes, one with lateral expansion only (Type B), one with both lateral and laid-back expansion (Type A), and a third with the addition of compound angle to the Type A shaping. The Type-A-shaped hole resulted in about a 10% lower loss coefficient than the Type B hole over the range of blowing ratios from 0 to 1.8 (a reduction in loss coefficient from 4% to 3.6%). Above a blowing ratio of 1.8, the hole shapes yielded the same results. The addition of a 45 deg compound angle served to increase the loss coefficient to the Type B level for blowing ratio up to 1, and drastically higher for greater blowing ratios, indicating mixing losses due to the turning of the injected coolant. Unlike the single film row test of [8], Day et al. [10] examined the mixing losses associated with wholesale changes in several turbine inlet vane film rows from round to shaped holes. Using a transonic vane cascade airfoil with 14 rows of film cooling on the airfoil (six showerhead/gill film rows, four pressure side rows, and four suction side rows), total changes in aerodynamic efficiency were measured with the last four film rows on both sides changed from round to shaped. The shaping was Type B with lateral angle of about 15 deg. The film row locations were typical of a highly cooled turbine inlet vane. They found the use of shaped film rows to essentially double the decrement in stage efficiency over that associated with the use of only round film holes (from −0.27% to −0.60% relative to the uncooled vane).

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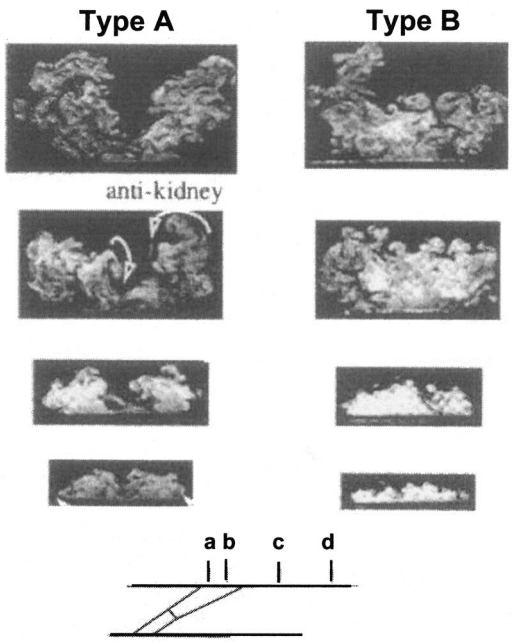

Fig. 9 Shaped film hole LIF images at transverse planes (reproduced from [6])

Fundamental Cooling Effects Due to Shaping

While it is an accepted fact that most shaped film holes provide higher centerline and laterally averaged film effectiveness than round holes given the same flow rate and hole spacing, there are surprisingly few published studies containing data on any one type of shaped geometry with similar parameters. For example, flat plate Type-A-shaped hole effectiveness experimental data listed in Fig. 4 is limited to the studies of [11,12], and [13], and even then parameters, such as L/D, spacing, and diffusion angle are different. Type-B-shaped hole effectiveness data is limited to the studies of [3,11,14], and [15]. One parameter not listed in Fig. 4 is that of the coolant-to-mainstream density ratio (DR). Among most film-cooling studies, DR takes on either very low values near 1.0, or more engine representative values approaching 2.0. Since DR has a strong effect on the resulting film effectiveness, this presents additional difficulties in comparing data sets. Studies are also typically divided into two subgroups depending on the extent of downstream data in terms of X/D, either near hole data with $X/D<20$, or full field data out to $X/D\sim100$. Observations about film-cooling parametric effects must then be drawn individually from studies on the basis of general and relative relationships, but not on the basis of absolute effectiveness magnitudes.

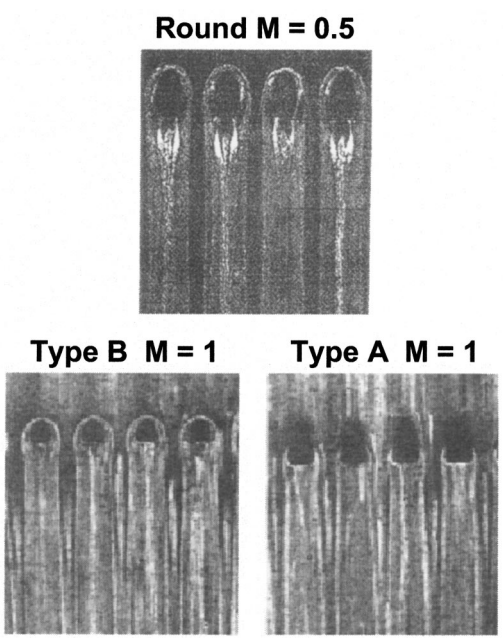

Fig. 10 Oil and dye surface traces for round and shaped film holes (reproduced from [7] and [8], respectively)

Blowing (Mass Velocity) Ratio. The basic effect of blowing ratio for shaped film holes has been shown in Figs. 6 and 7, from the study of [3], for a single geometry of film hole. But, do other shaped film holes perform in the same manner and magnitude? One of the more extensive investigations is that of Gritsch et al. [11] for both Type A and B shapes, using a DR of 1.85. This research is limited to $X/D\leqslant10$, focusing only on the developmental region of jet interaction. This data is also for a single film hole, however, the study of [3] showed little difference between single hole and row of holes results. Over the range of blowing ratio from 0.5 to 1.5, they find the Type B centerline film effectiveness to continuously decay downstream from peak levels of about 0.7, showing no signs of blowoff. Type A centerline effectiveness is very similar for the higher blowing ratios, but is severely degraded at $M=0.5$. The fan-shaped Type B holes resulted in higher centerline effectiveness at each blowing ratio compared to the laidback Type A holes. This improvement amounted to as much as 40–50% increases at $M=0.5$ and $M=1.5$, but only about 10% or less for $M=1$. In examining blowing ratios from 0.25 to 1.75, the fan-shaped Type B holes exhibited optimal effectiveness for blowing ratios of about 1–1.2, while the Type A holes showed no optimal blowing ratios, but instead increasing steadily as blowing ratio increased. From this study, it can be concluded that Type B film holes provide the highest centerline film effectiveness over the broad blowing ratio range up to about 1.5, but that Type A holes are more robust to very high blowing ratios. This latter observation is likely due to the greater diffusion volume provided by the laidback holes. The Type-A-shaped holes of both [12] and [13] also exhibit increasing centerline effectiveness levels as blowing ratio is increased over their more limited ranges. The Type B film hole data of [3], referring back to Figs. 6 and 7, show little sensitivity to blowing ratio changes from 0.5 to 2, with similar peak magnitudes to the data of [11]. In contrast, the investigation of Yuen et al. [14], using the same Type B geometry as in [11], but with a DR of 1.5, shows constantly increasing centerline and laterally averaged film effectiveness values as blowing ratio is increased from 0.33 up to 1.67. Laterally averaged film effectiveness is nearly 1 in this study at the highest blowing ratio, a result which has no independent verification as yet. The Type B data of Sargison et al. [15] is very much in agreement with that of [3] and [11].

Compound Angle (CA) Holes With Shaping. In the application of film cooling to turbine airfoils, it is generally preferred to orient holes in the axial direction to reduce mixing with the mainstream gases for better effectiveness, and to hold mixing losses to a minimum. There are regions of the airfoils, however, where the machining of axial holes would require the use of steep angles to the surface, perhaps as much as 60–70 deg, which would yield unacceptable cooling effectiveness. These regions are the high curvature areas of the aerodynamic profile on both the pressure and suction sides, typically in the forward portion of the airfoils, just aft of the showerhead radial film holes. These regions see the application of compound angle film holes, sometimes referred to as gill holes. The compound angle imposes both axial and radial components to the hole centerline relative to the surface tangent in the axial direction (or relative to the radial tangent,

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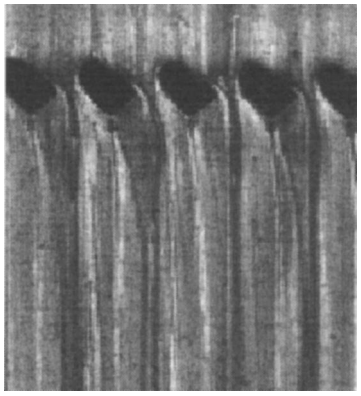

Fig. 11 Oil and dye surface traces for compound angle, shaped film holes (reproduced from [8])

though this is not the usual definition). The compound angle allows the hole centerline angle to the local surface to remain in the range of 20–40 deg for shallow injection and best effectiveness. Most Type-C-shaped holes using only diffusion in the laidback sense will be found in these regions, which is why there is little flat plate data on Type C holes. Only the study of [12] shows Type C data without compound angle on a flat plate, with results showing negligible improvement in effectiveness over round film holes. The research of Schmidt et al. [16] used a Type-C-shaped hole with a 60 deg radial compound angle on a flat plate surface and compared to a round hole of the same orientation. Their results showed the addition of the shaping to yield better uniformity of effectiveness laterally in downstream locations and to increase laterally averaged film effectiveness in the $X/D<10$ region by 20–50% for blowing ratios of 0.6 and 1.25, respectively. Both CA holes, shaped or not, showed improvements over the axial round hole effectiveness.

Shaping provides a much needed improvement to compound angle holes, albeit one that is difficult to manufacture with consistency. The natural tendency of the film injection from a CA hole is to lose much of its cooling potential in the process of turning and mixing with the mainstream flow. Shaping allows the film to remain closer to the surface for a more benign turning, one with lower mixing and greater film coverage. Figure 11 shows the oil and dye surface streamline traces obtained by Ganzert et al. [8] for a Type-A-shaped, 45 deg CA film hole with blowing ratio of 1.0. As noted above, the mixing losses are greater for this hole, but the traces indicate a very uniform and laterally dispersed coverage with turning confined to the hole exit region. A fundamental study of Type-C-shaped CA film holes by Chen et al. [17] investigated the application of such film rows to general concave and convex surfaces in a wind tunnel, with comparison to axial shaped holes. On a concave surface, the addition of compound angle of 45 deg made no improvement at low blowing ratio (0.5), but provided nearly a 30% improved laterally averaged effectiveness for $X/D<10$ at high blowing ratio of 2. On a convex surface, the addition of CA improved average effectiveness by 30% for the low blowing ratio, again in the $X/D<10$ region, and drastically improved effectiveness at high blowing ratio by preventing blowoff. The general benefit of hole shaping that was noted early is seen again here, that of more robustness in cooling effectiveness at high blowing ratios.

At least three studies have looked specifically at the airfoil leading-edge application of compound angle-shaped film holes. Reiss and Boelcs [18] investigated the use of both Type-B- and Type-C-shaped holes as improvements to the typically round compound angled holes in the stagnation showerhead region of the airfoil. They applied the changes to the stagnation film row and two additional showerhead film rows and tested two approach mainstream Mach numbers to represent vanes or blades. At the lower mainstream Mach number of 0.14, the Type C laidback film hole performed the best overall, especially at higher blowing ratios of 1 or greater, showing a 10–20% averaged effectiveness benefit over round CA holes. The Type B shaping showed in essence no improvement. At the higher mainstream Mach number of 0.26, the results are much more mixed. The Type C shaping does as well, and sometimes very much better than the round holes, but the improvements are much more localized to specific rows of film. Again, the Type B shaping shows little or no improvement. The research of Dittmar et al. [19] modeled a single row of gill holes using Type B shaping with and without compound angle. It is important to note that the tests of [19] used a small circular tube to supply the film rows rather than a large plenum, creating an additional internal coolant supply directional effect. The shaped film holes provided higher laterally averaged effectiveness in all cases, and particularly so at high blowing ratio. Improvements over round holes were 10–30%. The effect of compound angle on the shaped hole in this study was minor, and seen mostly in a less uniform distribution of cooling near the hole exit due to the internal supply mechanism. In a CFD study, McGrath et al. [20] show effectiveness predictions and details of flow for compound angled Type A holes on both pressure- and suction-side models (gill holes) and compare effectiveness to data obtained in a linear cascade of the same geometry. Laterally averaged effectiveness on the pressure side (concave) compared well with test data, showing effectiveness levels from peaks of 0.6 with well-behaved decay far downstream, similar to [17]. Increased blowing ratios from 1.5 to 4.5 resulted in increased effectiveness with no signs of blowoff. Similar results were found on the suction side (convex), but with much lower overall effectiveness levels for blowing ratios from 1 to 2, again similar to [17]. Axial-shaped hole data obtained by Ferguson et al. [21] for the same cascade exhibit lower film effectiveness on the pressure-side surface, showing signs of blowoff that the CA holes tend to resist. In contrast, axial shaped hole data on the suction side show significantly better film effectiveness than their CA-shaped counterparts, the latter being degraded from crossflow interaction effects and according to the CFD also hot gas ingestion.

Effect of Shaping on Heat Transfer Coefficients. In reviewing the various studies that have measured heat transfer coefficients with film injection, one key factor becomes immediately evident, that the resulting heat transfer coefficients are highly dependent on the state of the boundary layer approaching the film holes. This would appear to be an obvious point, and, in fact, the disturbance of the boundary layer, whether small or great, also depends greatly on the film hole blowing rate (velocity and momentum ratios to the mainstream), as well as the hole spacing or coverage in the direction transverse to the mainstream. Two main arguments can be made with respect to film injection effects, and either one can be true depending on the situation. On one end of the spectrum, a developed boundary layer, typically turbulent for cooled airfoils, will not be as disturbed by film hole injection as a thin or new boundary layer. At low blowing rates, or low coolant velocity relative to the mainstream, the film injection presents little effect on the turbulent boundary layer, and even serves to thicken it to the extent that the heat transfer coefficient is lower than that of the noninjected case. Only at sufficiently high blowing rates does the injection result in heat transfer coefficient enhancement over the undisturbed level. At the other end of the spectrum, a fresh or very thin boundary layer (relative to the film hole diameter) will feel the disturbance of the film injection for virtually all blowing rates, less for low blowing rate and greater for high blowing rate, but always yielding a heat transfer coefficient enhancement greater than unity. Such thin boundary layers may be found in the region of the airfoil leading edge and, generally, all along an airfoil that has significant numbers of film cooling rows (see Fig. 3) that serve to continuously disrupt the reforming boundary layer. The coverage of the film row, a somewhat loosely defined quantity representing the physical or fluid-

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dynamic percentage of lateral spatial injection, plays a role in determining if the injection disturbance is localized (low coverage) or total (full coverage). For example, shaped film holes with close spacing may have merging exit regions leading to 100% coverage, which in a thin boundary layer could cause a fresh start to the boundary layer. Research into the heat transfer coefficients for shaped film hole coolant injection tends to fall into these two categories of boundary layer flows, either injection into thick boundary layers or injection into thin boundary layers. The former cases may represent isolated film cooling rows for components receiving limited total cooling, for example, the second- or third-stage airfoils of the turbine. The latter cases are more representative of those experienced in most highly cooled turbine airfoils.

In the category of relatively thick boundary layers compared to the injection jet size ($\delta/D>0.5$), Makki and Jakubowski [22] measured the heat transfer coefficient (HTC) enhancement for shaped holes on a flat plate, as the ratio of HTC with film cooling to that without film cooling. For blowing ratios from 1 to 3 and a density ratio of 1.59, they observed a fairly constant HTC enhancement of about 0.8 compared to enhancement levels between 0.9 and 1 for round holes. These levels held out to nearly X/D of 100. At low density ratio of 1.15, there was more distinction of blowing ratio effect, showing enhancement of 0.9 at $M=0.9$ and increasing to 1.2 at $M=2.5$. Under this condition, most cases showed a rise or decay to enhancement approaching 1 downstream. In the flat plate study of Gritsch et al. [23], using a density ratio of 1.85, laterally averaged HTC enhancements of 0.8–1.2 were measured as blowing ratio increased from 0.25 to 1.75, with gradual change toward enhancement of 1 downstream. Their Type-B-shaped holes showed much less range of initial enhancement from 0.7 to 0.85, all rising toward enhancement of 1 at $X/D=10$ at essentially the same rate. Similarly for Type-A-shaped holes, the initial enhancements ranged from 0.6 to 0.75, again rising up to level at $X/D=10$. As a function of blowing ratio, the shaped holes exhibited a minimum HTC enhancement at $M\sim1.0$, with both lower and higher M leading to greater initial enhancements. The less extensive tests of Yu et al. [12] also showed initial enhancements of 0.7 rising to 1 at about $X/D=10$ for the blowing ratios of 0.5 and 1.0. The study of Sen et al. [24] employed a somewhat thinner boundary layer with $\delta/D\sim0.23$ at injection, with a density ratio of 1.6. Their HTC enhancement for a CA-shaped hole was about 1.4 initially with decay to about 1 at $X/D=25$, representing a higher and more prevalent effect.

Several studies have employed relatively thin boundary layers of $\delta/D\leqslant0.12$ at injection either through design or because of the model geometry and flow. Chen et al. [17] performed concave and convex wall-shaped film injection with density ratio of 0.98. They found HTC enhancements of 1.4–1.6 on the concave surface for $M=0.5$, which was the same as that of a round hole. Increasing the blowing ratio to 2, however, caused enhancement to rise to 1.5–2 or more, with levels persisting far downstream, especially for the CA-shaped holes. On the convex surface, enhancements ranged from 1.4 to 3 at $M=0.5$ as well as $M=2$, again with the higher blowing ratio case elevated far downstream. The airfoil leading-edge investigation of Reiss and Boelcs [18] with density ratio of 1.6, which naturally has a fresh boundary layer, showed mixed results of sometimes lower or higher HTC enhancements due to shaped holes, but generally averaging about a 1.2 factor overall. Dittmar et al. [19] obtained enhancements for shaped gill holes on a leading-edge model using density ratio 1.3. At low blowing ratio of $M\sim0.5$, HTC enhancements were 1.2–1.4, while at $M\sim1.5$ these were 1.4–1.8, and at $M\sim3$ values were from 1.2 to 1.6. In all cases the enhancements decayed to about 1 at $X/D\sim30$. In a blade cascade test, Ganzert et al. [8] observed shaped row enhancement of about 1.6 initially with only a limited decay for $M=1$ and 2. Because the suction-side injection caused the boundary layer to remain turbulent rather than transitioning to laminar flow, downstream enhancement was as much as 3 at $X/D=20$. In a vane cascade test with density ratio 1.7, Abuaf et al. [25] measured suction-side-shaped film HTC enhancement of 1.5, but in this case, decay was rapid to 1 at $X/D\sim20$. The vane cascade retained a turbulent condition even without film injection.

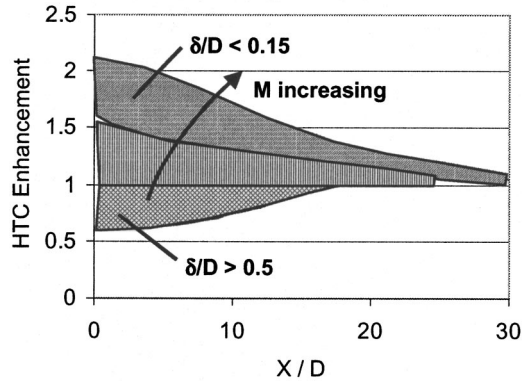

Fig. 12 Summary of heat transfer coefficient enhancements with shaped film holes

Figure 12 attempts to summarize the ranges of HTC enhancements observed for shaped film holes. The indicated regions may overlap, grow, or shrink depending on specific parameters, and there will always be circumstances that exceed these regions in terms of magnitude and extent. The studies noted employed film row hole spacing ranging from $2.5D$ to $5D$ based on the throat diameters, which with shaping will result in approximate geometric coverage values from 80% to 40%, respectively. The reported HTC enhancements results were by and large laterally averaged values such that any coverage effect is already included. One additional measure of film injection effect is the net heat flux reduction (NHFR), or simply the ratio of heat flux with film to that without film. This depends, of course, and can be derived from, the film effectiveness, the HTC enhancement, the gas and coolant temperatures, and the surface temperature, as per Sen et al. [24]. Film effectiveness is generally the dominant player in NHFR, with typical values of 0.8–0.5 just downstream of the film injection; but when injection disrupts the boundary layer severely, the NHFR can be greater than unity (i.e., higher heat flux with film cooling).

Effect of Film Hole Entry and Internal Conditions. Although much attention is paid to the outward results of adiabatic film effectiveness and heat transfer coefficients, relatively little notice has been given to the effects of the internal coolant feed mechanism or the interior of the holes themselves. It has become apparent in the last few years that effectiveness and heat transfer resulting from plenum-fed experiments do not necessarily represent true data for real airfoil cooling. This is especially true for film holes with length-to-diameter ratios of $\leqslant6$, which make up the majority of those used in practice. Issues of channel-fed film, internal channel flow velocity and orientation, resulting generation of turbulence within the film holes, and even internal surface features of channels and film holes can all significantly affect the external cooling. Gritsch et al. [26] assessed the discharge coefficients for round and shaped film holes under various conditions of internal channel Mach number as well as parallel or perpendicular cross flow feeding the film holes. Drastic reductions in discharge coefficients were observed for all hole types as the internal channel Mach number was raised from 0 to 0.6. Similar large reductions were noted when film holes were fed by perpendicular cross flow rather than parallel cross flow for a given internal channel Mach number. Likewise, the work of Bunker and Bailey [27]

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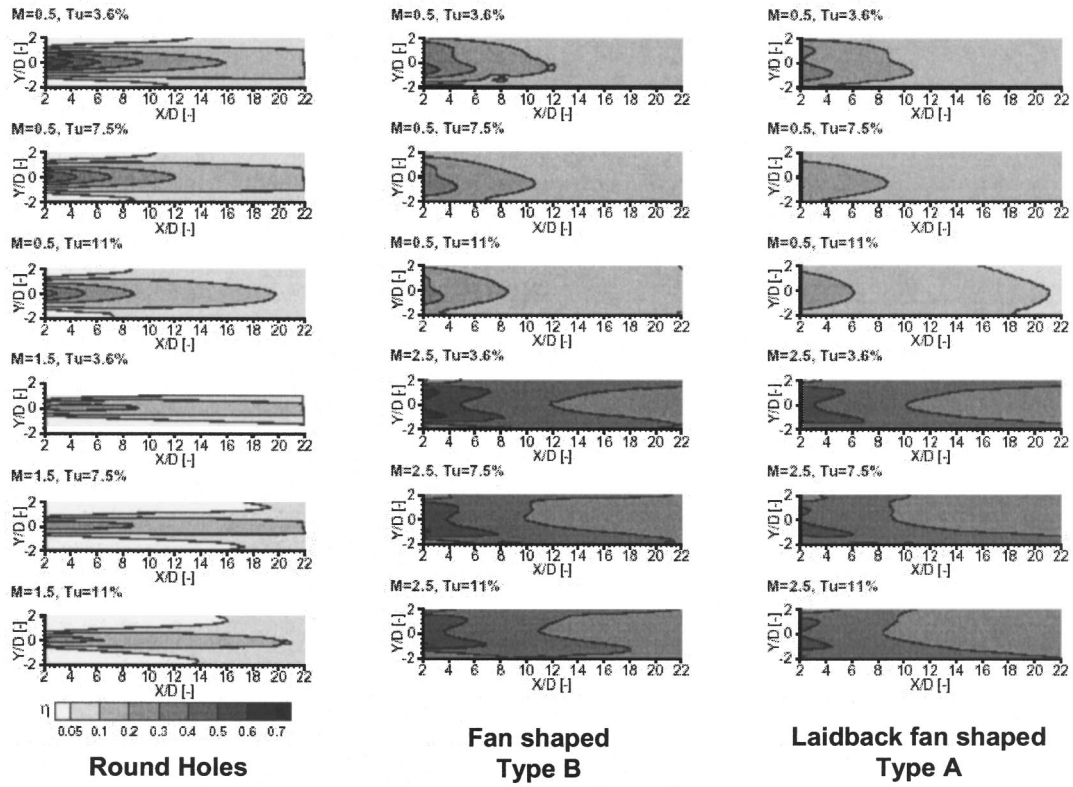

Fig. 13 Effect of freestream turbulence intensity (reproduced from [31])

showed significant discharge coefficient changes due to the relative placement of film hole entries near internal channel turbulators.

Carrying such effects into the associated film effectiveness and heat transfer measurements, Gritsch et al. [11] showed that increased internal channel Mach number up to 0.6 served to increase round hole laterally averaged effectiveness by about 30% for the region of $X/D<10$, but had impacts in decreasing near-hole responses for shaped holes by 15 to 20% at $M=1$. Gritsch et al. [23] showed the companion effects on heat transfer coefficients of about 10% increase for round holes, 15% increase for Type-B-shaped holes, and 30% increase for Type-A-shaped holes. Both studies used parallel coolant channels to the hole entry. Saumweber et al. [28], also using $M=1$, showed the effect of perpendicular internal channel cross flow on Type-B-shaped holes to be a 10–20% reduction in effectiveness compared to parallel feed, with a skewing of the local surface effectiveness distribution due to the altered flow into the holes. The CFD study of Kohli and Thole [29] came to the same conclusions regarding the effects of internal channel Mach number and cross flow on shaped holes and showed even stronger effects at $M=2$. This and other CFD studies, such as those of [20] and [21], provide predictions of the detailed flow within shaped holes showing entry jetting and recirculation regions affecting the ultimate issue of the film cooling. In the more specific case of a tube supply chamber feeding a row of shaped gill film holes, Dittmar et al. [19] found that the direction of coolant flow within the supply tube drastically affected both the film effectiveness and heat transfer coefficients for CA-shaped holes. Supply of the coolant in the direction leading to a shallow angle of entry into the film holes resulted in 25–50% higher effectiveness and also somewhat higher HTC than supply requiring a greater than 90 deg turn into the holes. The interaction of coolant supply means with the film holes and their shapes is obviously a very complex process with nearly unlimited possible configurations, but also potential for advantage in performance. Finally, while film holes are today manufactured and coated by several means, resulting in differing internal surface morphologies, there is so far only one published study looking at the change in film effectiveness due to any such interior surfaces. Bunker [13] examined the effect of partial coating blockage on film effectiveness for round and shaped holes. For EDM film holes with actual plasma-sprayed protective coatings applied as in a land-based power turbine, the potential for severely reduced effectiveness is present in round holes with as much as 50% loss. Shaped holes of Type A were found to much more tolerant of partial blockages from applied coatings.

Effect of Freestream Turbulence (Tu) Level. The majority of the aforementioned studies into shaped film holes utilized relatively low freestream turbulence intensities, generally less than 4%. Research into the effects of freestream turbulence on round film hole effectiveness has shown that elevated intensity levels can quite severely reduce effectiveness. The investigation of Saumweber et al. [30] presents both effectiveness and heat transfer coefficient enhancement data for Type A and B shaped holes with variations in freestream turbulence intensity from 3 to 11%. Figure 13 shows the detailed effectiveness distributions downstream of both round and shaped holes for low and high blowing ratios. This data also depicts many of the general differences between round and shaped hole effectiveness noted earlier. With regard to the effect of Tu at low blowing ratio, round holes experience a 20–40% reduced effectiveness as Tu is increased, while shaped holes show only a 10–20% reduction. At high blowing ratio, there is very little effect of Tu to be seen in round or shaped holes, as the jet strength and jet turbulence levels dominate the responses. The two shaped-hole types show essentially the same effectiveness under similar conditions. The HTC enhancements for round and shaped holes with low Tu are those noted in [23] and discussed above. At high free-stream Tu levels, enhancements for round holes increase for all blowing ratios to about 1.1. En-

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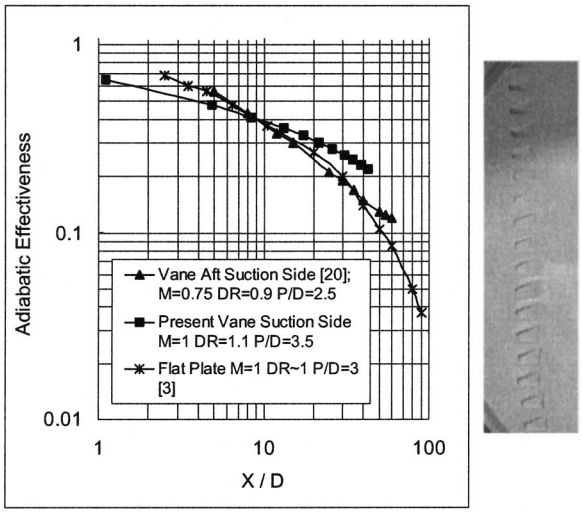

Fig. 14 Cascade suction side film effectiveness with shaped film (present test surface photo)

hancements for shaped holes at high blowing ratios of $M=1.5$ and 2.5 also increase with Tu, but much higher to about 1.7 initially, followed by decay to about 1.4 by $X/D=20$. Under low blowing ratio of $M=0.5$, this response is lessened to that of the round holes. Shaped film hole injection effectiveness shows less sensitivity to freestream turbulence levels, though reduction is observed, but also demonstrates a substantial magnification of the heat transfer coefficients.

Cascade Shaped Film Data

The primary purpose in using shaped film holes is the resulting increase in adiabatic film effectiveness over that obtained with round film holes. The database of literature concerning shaped film effectiveness rests mostly on flat plate experiments, but is this data useful for design? There have been several airfoil cascade tests using shaped film holes for one or more film rows, but most of these investigations do not show the isolated effectiveness due to shaped film holes in the cascade environment, making it difficult to assess individual row effectiveness. Two cascade tests do provide data on a single shaped film row on the suction side of a vane. Takeishi and Aoki [31] provide data for a row of Type-A-shaped holes midway down the suction side issuing prior to the passage throat in a low-speed cascade. Their airfoil was made of a low-conductivity insulating material. Recent cascade test data from the present author also utilized a single row of Type-A-shaped film holes placed midway on the suction side prior to the throat, but in a high-speed transonic cascade. This cascade used an actual production heavy frame turbine airfoil with film holes machined by standard methods. Figure 14 shows the centerline film effectiveness for both cascades, as well as the shaped hole data of [3] at a similar blowing ratio of $M=1$. Also shown is a close-up view of the film row tested in the present research. It is notable that all three film curves shown agree fairly well despite the differences in test environments. This is not altogether unexpected though, since the suction side of the airfoil will see an effective low free-stream turbulence intensity. The use of a single film row in such a location also means that the boundary layer will be thicker than otherwise expected if typical upstream film row injections were present to disrupt and restart the boundary layer. These factors, in essence, level the playing field between the tests and result in good agreement. The main point behind this comparison is that fundamental film effectiveness data can be highly useful for turbine design purposes when it is performed under compatible enginelike conditions for those parameters most affecting film-cooling performance. Under noncompatible conditions the data are not of design use, but rather point to relative results that might distinguish between hole geometries.

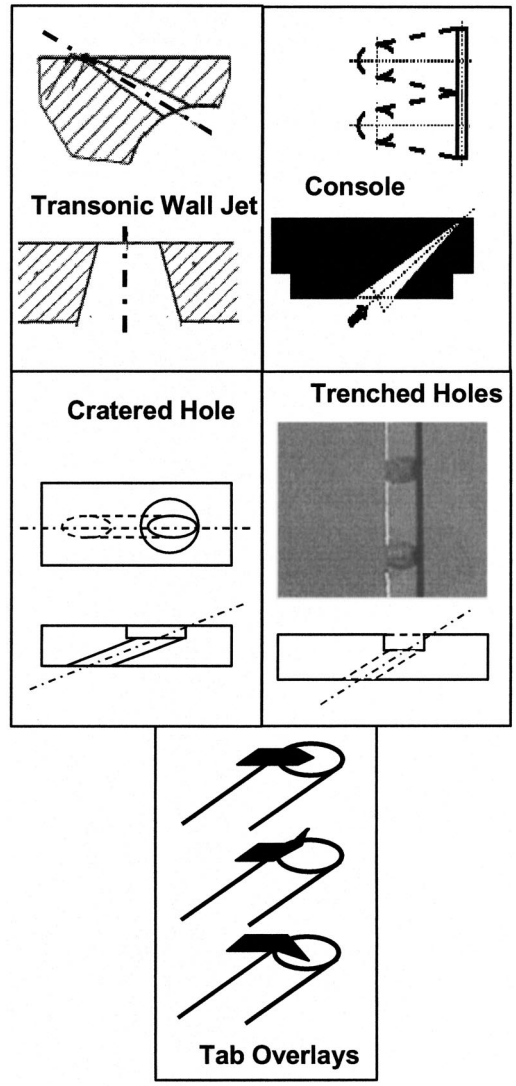

Fig. 15 Proposed alternative film holes shapes

Special Shapes Shown to Improve Effectiveness

Several alternate geometries of film-cooling holes have been proposed within the last few years that have in some form demonstrated at least equivalent film effectiveness performance to the shaped holes discussed above. These differing film holes may have specific form and function, either of limited or widespread potential, but each must also ultimately face the challenges of manufacturing, operability, and cost effectiveness. Figure 15 shows five such geometries that will be briefly explained here.

Moser et al. [32] analyzed a transonic wall jet geometry in which the film hole transitions from some interior cross section, perhaps circular, to a surface slit. This transition is such that the hole converges in both the axial and lateral directions. The basic principle of this design is the creation of a choked flow and underexpanded jet that will conform to the exterior curved surface via the Prandtl-Meyer effect for expansion waves. Their analysis shows that such film jets are very insensitive to angle and show no lift-off from the surface. Because the jet is underexpanded, the local film effectiveness according to the conventional definition

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exceeds unity at the exit. Such a jet requires a heavy pressure ratio to be realized, however. The analysis did not evaluate aerodynamic losses.

Sargison et al. [33] and [34] demonstrated a converging slot-hole, or Console, geometry in both flat plate and cascade testing. In this geometry, the hole transitions from circular to slot with convergence in the axial direction and divergence laterally. The hole area does diminish to cause the flow to accelerate, though not necessarily to a choked condition. The exit is the metering section. This accelerated flow is speculated to have lower jet turbulence and more stability. Flat plate tests show the effectiveness and HTC enhancement to be on par with fan-shaped holes, while the aerodynamic loss is less. Transonic cascade tests comparing Console to fan-shaped holes showed higher effectiveness and HTC on the accumulated suction-side film cooling from multiple rows and about equivalent effectiveness with higher HTC on the pressure side accumulated film. The cascade aerodynamic losses were much decreased.

Fric and Campbell [35] investigated a so-called cratered film hole in which the circular hole exits into a shallow right circular surface cup or depression. The flow actually impinges on the edge of this depression causing it to deflect and fill the depression prior to issuing onto the external surface. Flat plate tests show about a 50% improved effectiveness over round holes at $M=1$ and a greater increase as blowing ratio increases. Effectiveness improvement of 100% and more was observed at $M=5$. A suggested application in [35] is the full coverage film cooling of combustor liners.

Bunker [36] extended the work of [35] to a two-dimensional version, or trench film geometry. The intention of this geometry is the same, to cause the film-cooling flow to spread into the trench prior to issuing on the surface. The particular advantage of this geometry is that it can be formed using the protective coatings applied to the surface, without machining the trench into the substrate. Film effectiveness improvements of 50–75% over round holes were measured in flat plate tests, putting this on par with shaped holes.

Finally, Nasir et al. [37] tested round film holes with the addition of triangular tabs covering the upstream edge of the holes, either turned upward, downward, or flat. The flat and downward tabs modified the exit flow and vortex structure in a manner that kept the coolant from lifting off. Their data showed effectiveness consistent with shaped holes, but much higher HTC enhancement level of about 2.

Conclusions

Shaped hole film-cooling technology has become the standard in today's highly cooled gas turbine airfoils. This review has examined the origins of shaped film cooling and summarized the current state of knowledge concerning this technology, including the flow field, heat transfer coefficients, and adiabatic film effectiveness. The benefits of shaped film holes using expanded exits of fan and/or laidback geometries may be summarized as follows:

- higher centerline and laterally averaged adiabatic film effectiveness, in some cases approaching that of two-dimensional slot film
- little variation in effectiveness over blowing ratios from 0.5 to 2
- virtually no susceptibility to film blow-off under typical conditions
- continuously increasing effectiveness with blowing ratio
- low variability in heat transfer coefficient enhancement with blowing ratio
- low sensitivity to free-stream turbulence intensity variations
- amenable to use with compound angles

Shaped film holes also have some drawbacks, however, including higher aerodynamic mixing losses for the most common regions of use and more sensitivity (negative) to variations in entry flow orientations and strengths.

Of the 2500+ publications dealing with film cooling, only 50 or so specifically examine shaped, expanded exit film hole aerodynamics and thermal performance. The current understanding of shaped film parametric effects and physics is relatively sparse, however, the acceptance of shaped film technology is based largely on the foundation of round film hole data. The benefits of shaped hole film cooling are real and substantial, so much so that these types of film holes are used whenever possible in the practice of cooling gas turbines. No single shaping of film hole stands out as an optimal geometry, as long as the general guidelines of diffusion angles and sufficient depth/length of transition are followed. As manufacturing methods progress further, additional unconventional shapes of film holes will become viable, possibly leading to even higher performance and the ability to cool components with less air.

Nomenclature

D = film hole throat internal diameter
DR = density ratio $=\rho_{jet}/\rho_{gas}$
I = momentum flux ratio $=\rho_{jet}V_{jet}^2/\rho_{gas}V_{gas}^2$
L = film hole length
L_T = film hole throat section length
M = blowing ratio, $=\rho_{jet}V_{jet}/\rho_{gas}V_{gas}$
P = spanwise pitch between film hole centerlines
PR = film hole pressure ratio=Coolant total/Mainstream static pressure
Re = jet Reynolds number based on film hole diameter, $\rho_{jet}V_{jet}D_{jet}/\mu_{jet}$
$T_{coolant}$ = film hole exit static temperature
T_{gas} = freestream gas temperature prior to film row
$T_{recovery}$ = local recovery temperature of gas
$T_{surface}$ = local surface temperature with film cooling
Tu = mainstream turbulence intensity
V_{jet} = velocity of film jet based on throat diameter
V_{gas} = velocity of freestream gas
x = axial distance measured along surface from film hole exit
z = lateral distance from film row centerline
δ = boundary layer thickness
η = adiabatic local film cooling effectiveness, $=(T_{recovery}-T_{surface})/(T_{recovery}-T_{coolant})$
μ_{jet} = dynamic viscosity of film jet at exit
ρ_{jet} = density of film jet at exit
ρ_{gas} = density of freestream gas

Acronyms.

CA = compound angle
HTC = heat transfer coefficient
LA = laterally averaged value
LDV = laser Doppler velocimetry
PIV = particle image velocimetry
PLIF = planar laser-induced fluorescence

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